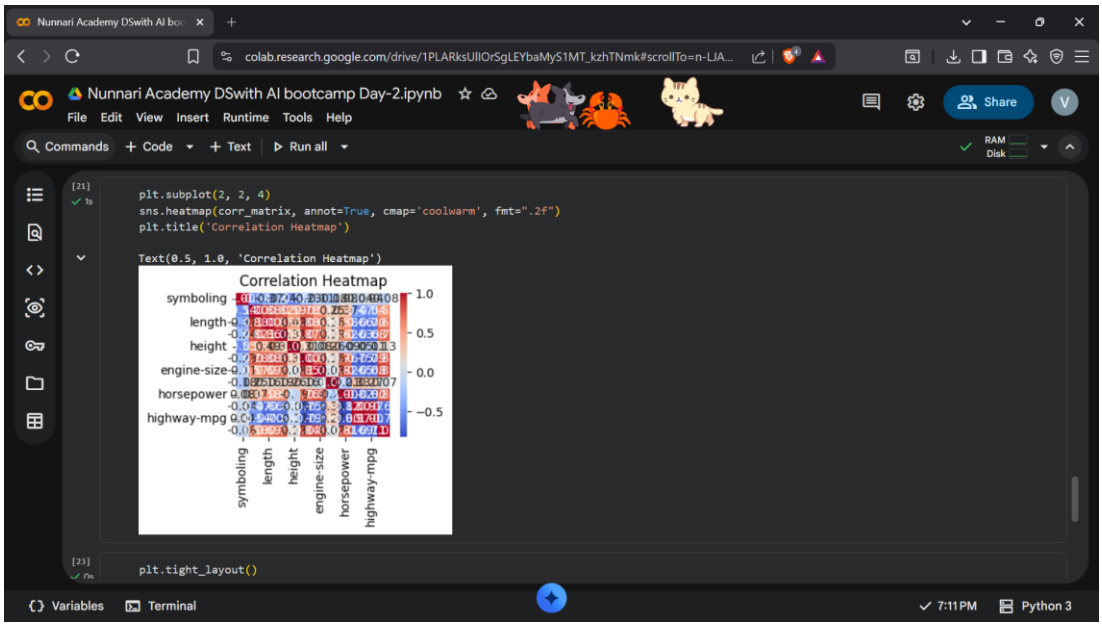
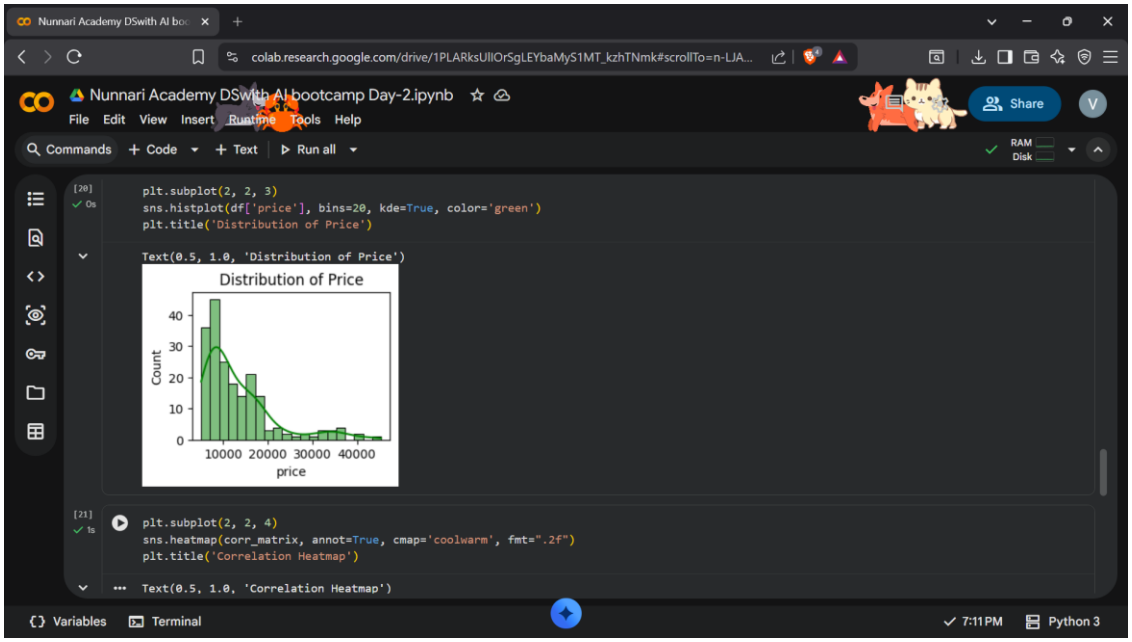
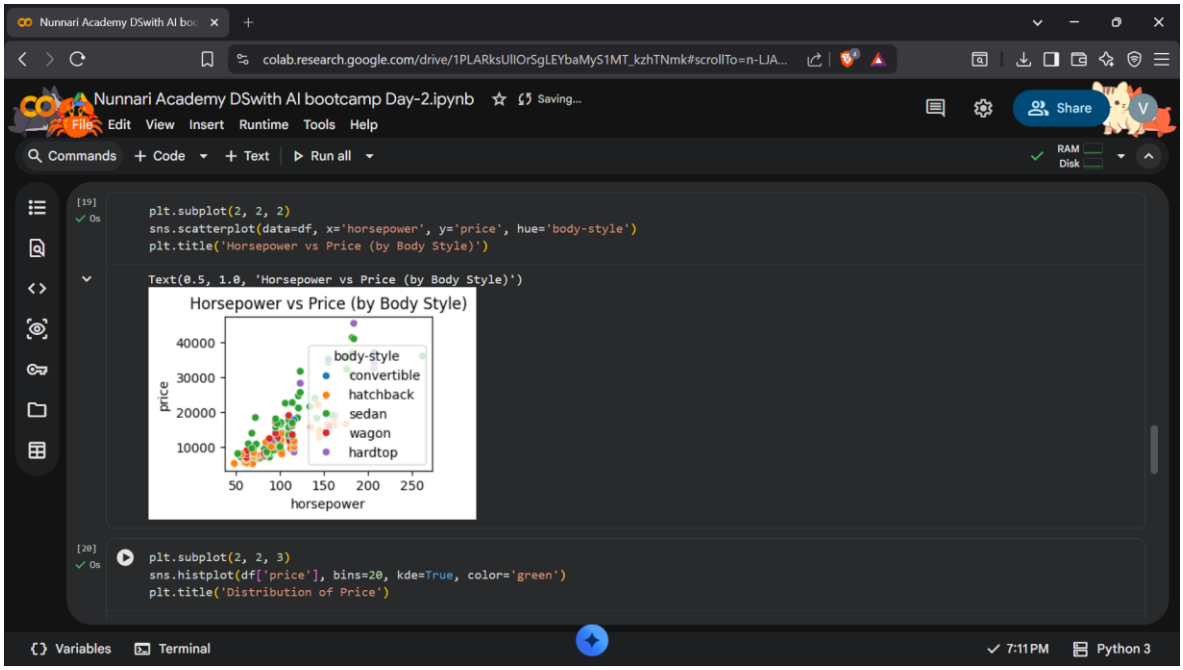
Nunnari Academy Datascience with AI Bootcamp

DAY-3 work Exercises



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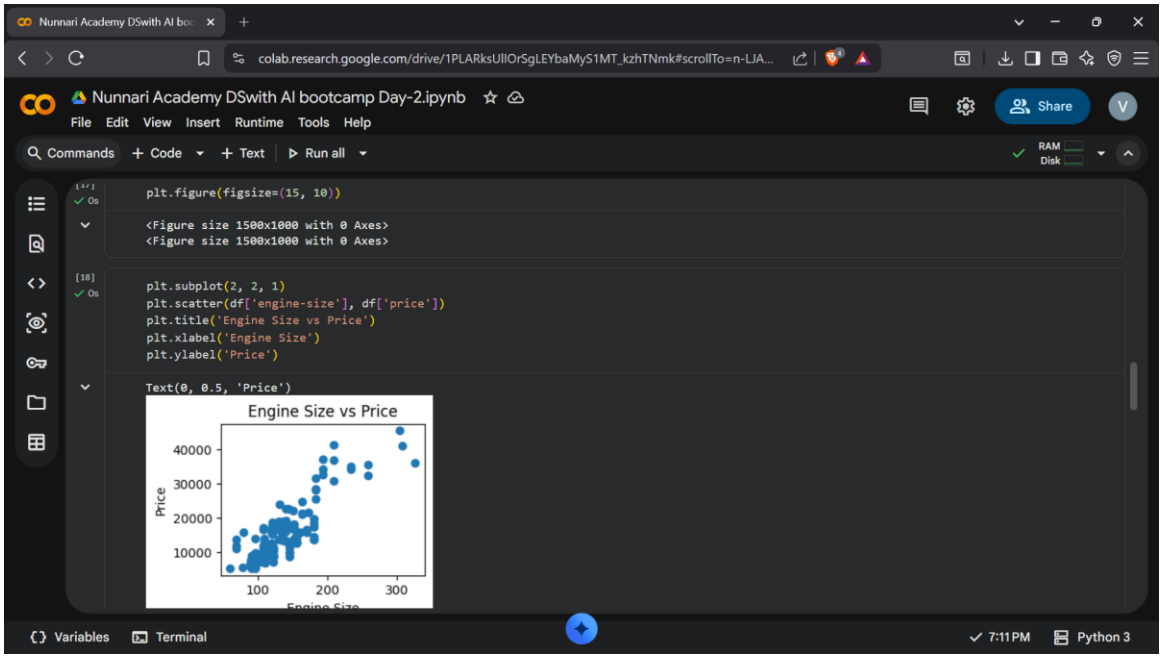
[19]
plt.subplot(2, 2, 2)
sns.scatterplot(data=df, x='horsepower', y='price', hue='body-style')
plt.title('Horsepower vs Price (by Body Style)')
Text(0.5, 1.0, 'Horsepower vs Price (by Body Style)')
Horsepower vs Price (by Body Style)
price
horsepower
body-style
convertible
hatchback
sedan
wagon
hardtop
[20]
plt.subplot(2, 2, 3)
sns.histplot(df['price'], bins=20, kde=True, color='green')
plt.title('Distribution of Price')
Variables Terminal
7:11 PM Python 3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plt.figure(figsize=(15, 10))
<Figure size 1500x1000 with 0 Axes>
<Figure size 1500x1000 with 0 Axes>
[18]
plt.subplot(2, 2, 1)
plt.scatter(df['engine-size'], df['price'])
plt.title('Engine Size vs Price')
plt.xlabel('Engine Size')
plt.ylabel('Price')
Text(0, 0.5, 'Price')
Engine Size vs Price
Price
Variables Terminal
7:11 PM Python 3

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[15]

```python
style_stats = df.groupby('body-style')['price'].agg(['min', 'max', 'mean'])
print("\nBody-style Price Stats:\n", style_stats)
```

```
Body-style Price Stats:
                  min      max          mean
body-style
convertible  11595.0  37028.0  21890.500000
hardtop       8249.0  45400.0  22208.500000
hatchback     5118.0  22018.0   9958.373134
sedan         5499.0  41315.0  14459.755319
wagon         6918.0  28248.0  12500.166667
```

[16]

```python
numeric_df = df.select_dtypes(include=[np.number])
corr_matrix = numeric_df.corr()
top_3_price = corr_matrix['price'].sort_values(ascending=False).iloc[1:4]
print("\nTop 3 columns correlated with price:\n", top_3_price)
```

```
Top 3 columns correlated with price:
 engine-size    0.873887
curb-weight    0.835090
horsepower     0.810533
Name: price, dtype: float64
```

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[12]

```python
df['price'] = pd.to_numeric(df['price'], errors='coerce')
df['horsepower'] = pd.to_numeric(df['horsepower'], errors='coerce')
df.dropna(subset=['price', 'horsepower'], inplace=True)
```

[13]

```python
print("\nMost common brand:", df['make'].value_counts().idxmax())
print("Most common body-style:", df['body-style'].value_counts().idxmax())
fwd_pct = df['drive-wheels'].value_counts(normalize=True).get('fwd', 0) * 100
print(f"Percentage of fwd: {fwd_pct:.2f}%")
```

```
Most common brand: toyota
Most common body-style: sedan
Percentage of fwd: 58.29%
```

[14]

```python
avg_price_make = df.groupby('make')['price'].mean()
print("Most expensive brand:", avg_price_make.idxmax())
```

```
Most expensive brand: jaguar
```

[15]

```python
style_stats = df.groupby('body-style')['price'].agg(['min', 'max', 'mean'])
print("\nBody-style Price Stats:\n", style_stats)
```

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```
[9]
print("Missing values check:\n", df.isnull().sum().sum())

Missing values check:
 0

[10]
cols_to_drop = ['Cabin', 'Ticket', 'PassengerId']
df.drop(columns=[c for c in cols_to_drop if c in df.columns], inplace=True)

[11]
rename_dict = {
    'Pclass': 'Passenger_Class',
    'SibSp': 'Siblings_Spouses',
    'Parch': 'Parents_Children'
}
df.rename(columns=rename_dict, inplace=True)
print("\nColumns after rename:", df.columns.tolist())

Columns after rename: ['symboling', 'normalized-losses', 'make', 'fuel-type', 'aspiration', 'num-of-doors', 'body-style', 'drive-wheels', 'en

[12]
df['price'] = pd.to_numeric(df['price'], errors='coerce')
df['horsepower'] = pd.to_numeric(df['horsepower'], errors='coerce')
df.dropna(subset=['price', 'horsepower'], inplace=True)
```

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Data visualisation and Heatmaps

```
[1]
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

[8]
url = "/content/sample_data/automobile_data.csv"
df = pd.read_csv(url)

[9]
if 'age' in df.columns:
    df['age'] = df['age'].fillna(df['age'].mean())
if 'embarked' in df.columns:
    df['embarked'] = df['embarked'].fillna(df['embarked'].mode()[0])

print("Missing values check:\n", df.isnull().sum().sum())

Missing values check:
 0
```

The order might be wrong but I have executed all the code and the solutions with it.

Thankyou.